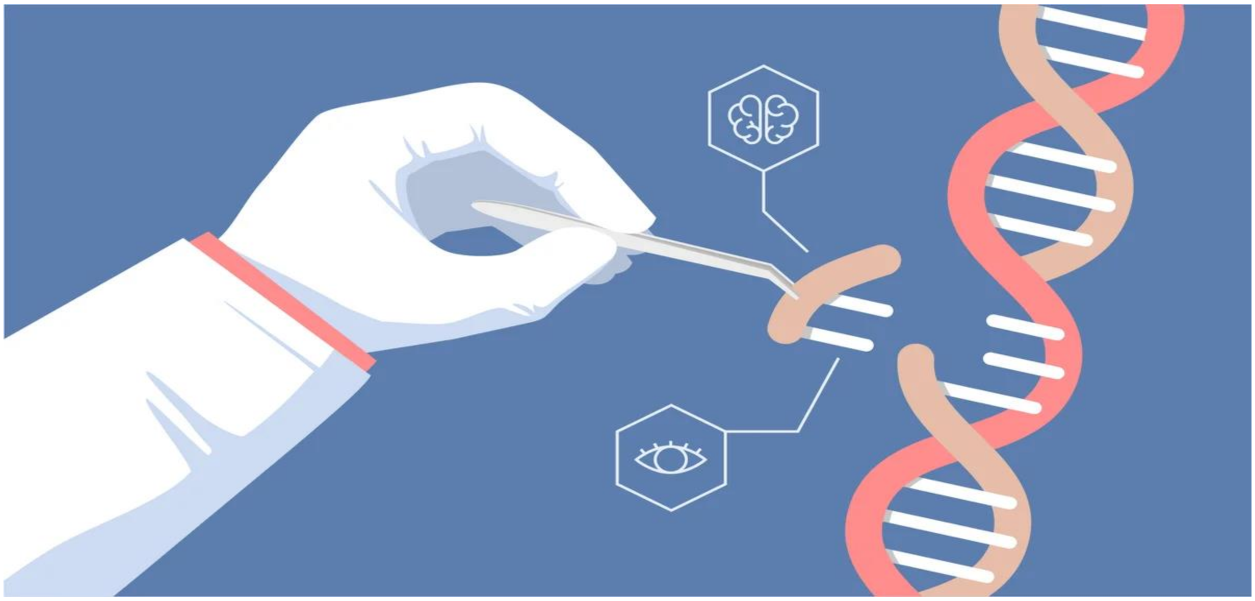
“Editing the Genome: Scientific Promise and Moral Responsibility”



DNA is the central theme of life, the molecule harboring genetic information for all living and some viruses. As per the central dogma of molecular biology, genetic information moves from DNA to RNA to protein via transcription and translation. Both RNA and DNA work together to ensure proper protein synthesis because one error can lead to dangerous diseases. Mutations are permanent changes in the DNA sequence which can be harmless or fatal, sometimes impairing vital genes or interfering with proper cell growth.

Now imagine having the ability to fix these errors in DNA like a copy editor, so that the genetic writing looks perfect. CRISPR-Cas9 is one such innovation of the 21st-century . CRISPR refers to 'clustered regularly interspaced short palindromic repeats ‘. CRISPR is a microbial immune system that protects against viral infections. It uses the Cas9 protein to recognize and cut specific DNA sequences of invaders. By supplying a guide RNA, scientists can program Cas9 to target and cut any desired DNA sequence. Since 2012, the gene editor has revolutionized biological research so that disease can be studied by scientists more readily and drugs can be developed faster. The technology has also revolutionized crop, food, and industrial fermentation design.

In agriculture, CRISPR can be used to develop crops with greater yields or that are resistant to drought, far more rapidly than is currently possible with traditional breeding methods. It can also be used to add new traits to crops or to remove others — such as creating gluten-free wheat or decaf coffee beans.

Other than environmental and agricultural applications, the most promising application of CRISPR is in health. The one application that has made it famous is editing the human genome, which promises the use of CRISPR in disease treatment. Gene therapy can mean two things: One is the repair of a broken gene, and the other is the regulation of the expression of a gene into protein products.

Our current understanding of gene therapy is ongoing and constantly developing. For example, sickle cell anemia is a disease that is normally caused by one mutation. So we can program CRISPR to fix it. Cell therapy is a method of engineering a patient's T cells to recognize and kill tumor cells more efficiently, for example, in leukemia. However, such modified cells can sometimes harm healthy cells or be dysfunctional. CRISPR improves their precision and safety so that treatments can be more efficient and regulated.

Yet the therapeutic application of CRISPR also recalls ethical concerns. *One fundamental ethical principle is not to do harm.* Some of the morally questionable areas are disease prevention and elimination of pest populations, and a few clearly immoral areas are enhancement and designer babies. The idea of "designer babies." is unethical because germ cell editing, i.e., egg or sperm, could possibly modify not just one individual but also their descendants, and this could lead to a new human species. Gene therapy is ethically problematic when applied to curing, preventing, and improving human traits. Cures are acceptable and welcomed, but prevention and enhancement are socially and morally tricky, particularly on the basis of justice and necessity. CRISPR applications for non-clinical purposes, for example, eradicating mosquitoes, could be harmful to ecosystems. Therefore, strong legislation and regulations are necessary to regulate this potent technology with so many possibilities.

In the end, CRISPR is a scientific promise and ethical dilemma — a technology that can transform the face of medicine and life itself, but only if utilized with wisdom, ethics, and responsibility.

REFERENCES:

1. *The CRISPR Revolution*. (2025, January 21). National Institutes of Health (NIH). https://www.nih.gov/about-nih/nih-turning-discovery-into-health-/transformative-technologies/crispr-revolution
2. *What is CRISPR? A bioengineer explains*. (n.d.). Stanford University. https://news.stanford.edu/stories/2024/06/stanford-explainer-crispr-gene-editing-and-beyond
3. Adam, J., & Adam, J. (2025, January 15). *CRISPR-Cas9: The gene editing tool changing the world*. Labiotech.eu. https://www.labiotech.eu/in-depth/crispr-cas9-review-gene-editing-tool/